MELODIUS

Twin Piano Experience

Progress Presentation · Group 6 GPP

🔧 PROTOTYPE PHASE

1. Prototype Demo
2. Target Users
3. Challenges & Plan B
4. Next Steps

WHERE WE ARE NOW and Prototype Demo A

✓ Concept & Design

✓ System Architecture

✓ Single Piano Build

✓ IoT Comms Setup

▶ Twin Sync Logic

○ Enclosure & Finish

✅ Completed

PLAN B

- ESP32 microcontroller programmed
- Push-button key detection working
- MQTT messaging over WiFi established
- LED feedback synced to key press
- Sound playback on both piano units
- Two-unit communication tested locally

✅ Stable & Working

LED + Sound Synchronisation

- MQTT event triggers LED glow on matching key
- Audio plays instantly on receiving piano
- Lower power, more robust over internet
- Visually intuitive , light guides the experience
- Ready for demo if solenoid not resolved in time

WHERE WE ARE NOW and Prototype Demo B

✓ Concept & Design

✓ System Architecture

✓ Single Piano Build

✓ IoT Comms Setup

▶ Twin Sync Logic

○ Enclosure & Finish

🔧 **In Progress / Testing**

PLAN A

- Solenoid actuator wiring & testing - completed
- Hall sensor / magnet key detection - completed
- Twin synchronisation latency tuning
- Enclosure 3D model (design phase for solenoid)
- Enclosure 3D Printing for both
- Battery & power integration

🔬 **Finalising and Actively Testing**

Solenoid Mechanical Key Sync

- Solenoid plunger physically presses keys on Piano B
- Servo / solenoid wiring tested on breadboard
- Physically mirrors Piano A , visual & tactile sync
- Challenge: latency & power draw being evaluated
- Currently exploring timing calibration

ENCLOSURE DESIGN

Fusion 360 · 3D Model Complete

✅ READY TO PRINT

METHOD
FDM 3D Printing

TOOL
Autodesk Fusion 360

KEYS
8-key layout

STATUS
Awaiting print

Melodius_Enclosure_v1.f3d — Fusion 360

TARGET USERS & SCENARIOS

Family Members

Family Connection

Separated by distance, they share a musical moment .A family member plays and another sees and hears the same notes light up in real time.

Long-Distance Couple

Romantic Bond

Partners in different cities share an intimate musical exchange, a melody becomes a love letter sent through sound.

Friends / Pen Pals

Playful Bonding

Two friends rediscover play; no music knowledge needed. Pressing keys together creates spontaneous joy across any distance.

No musical experience required — Melodius is about connection, not performance.

CHALLENGES & MITIGATIONS

Network Latency

Problem: Real-time solenoid sync is highly sensitive to WiFi jitter , any delay over ~80ms breaks the illusion of simultaneous play.

Mitigation: fallback to LED+sound (Plan B) which tolerates minor delays gracefully.

NEXT STEPS

This Week

1. Finalise solenoid decision go/no-go by end of sprint
2. Confirm Plan B LED+Sound as reliable demo fallback
3. Confirm key notes/melody
3. Refine MQTT latency benchmark over public internet
4. Begin Printing 3D enclosure for Twin Pianos
5. Integrate battery + power management (sleep modes)

Next 1 Week

1. Small user test: emotional response from non-musicians
2. Melodius video shooting

Final Sprint

1. Refinement & stability testing (stress test over distance)
2. Polish UX: LED effects, sound mapping, key feel
3. Demo prep, documentation, and presentation finalisation

Feedback welcome — we are open to suggestions on solenoid vs LED approach and any scope changes.

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Connecting through Music

Questions & Feedback

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